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Efficient Facial Emotion Recognition Using An Optimized Deep Learning Model Based On Quantum Gazelle Optimization Algorithm

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Abstract

This study proposes a new approach for the Facial Expression Recognition (FER) system that combines the Quantum Gazelle Optimization Algorithm (QGOA), local binary patterns (LBP), and Histograms of Oriented Gradients (HOG) with an optimized Deep Neural Network (DNN) classifier. The system uses computer vision techniques and deep learning algorithms to identify emotions in facial expressions. The proposed technique first employs the HOG and LBP descriptors, crucial components with excellent pattern recognition capabilities. These descriptors provide features resilient to small local changes in posture and lighting. However, they also generate unimportant and obtrusive characteristics that hinder classification performance. The proposed approach uses a wrapper-based feature selector called QGOA to solve this issue, which decreases the training complexity and improves recognition performance. QGOA takes advantage of the properties of quantum computing to regulate the diversity of face features and make proper selections using quantum measurements and Q-bit superstitious states. Finally, the optimized DNN detects facial emotions based on the selected features. The proposed approach was tested on the widely adopted FER2013 dataset. The results of the extensive analysis demonstrate the effectiveness of the proposed approach over state-of-the-art systems.

Keywords: FER, Deep learning, Gazelle optimizer, quantum computing, DNN, HOG, LBP, Feature selection

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1. Introduction

Emotion recognition, also called affective computing, represents a critical field of research at the intersection of artificial intelligence (AI) and psychology. This interdisciplinary domain seeks to enable computers to interpret human emotions, traditionally communicated through various means such as facial expressions, vocal nuances, and body language. The fundamental objective is diminishing the emotional disconnect between humans and machines, fostering more intuitive human-computer interactions.

Among the diverse modalities for emotion detection, this paper primarily focuses on facial expression analysis. Facial expressions are universally acknowledged as a potent and instinctive means of nonverbal communication, playing an instrumental role in human social interactions. According to [1], a substantial portion (55%) of emotional information is conveyed through nonverbal channels, predominantly facial expressions. These expressions serve as a window into an individual's emotional state, offering insights into their feelings and intentions.

The theoretical framework for understanding facial expressions has been subject to extensive research. One perspective, as delineated by [2], posits the existence of fundamental emotion levels, namely valence and arousal, which underpin all other emotional states and manifest through facial expression variations. In contrast, Ekman and Friesen's theory [3] identifies seven basic emotions—anger, happiness, disgust, fear, sadness, and surprise—that are universally represented through distinct facial expressions. They introduced the Facial Action Coding System (FACS), a comprehensive method based on the analysis of facial muscle movements. Despite its widespread adoption in psychological and neurological studies, FACS has limitations in capturing the full complexity and nuances of emotions encountered in daily human experiences. With the evolution of technology, automated facial expression recognition (FER) has emerged as a prominent area of research. Pioneering efforts in automated FER, employing computer vision and statistical analysis, were initiated as early as 1994 [4]. However, early techniques, predominantly reliant on manual feature extraction and traditional machine learning algorithms, were constrained by their inability to adequately handle the intricate and variable nature of facial expressions and the vast datasets involved.

A paradigmatic shift in FER has been witnessed with the advent of deep learning methodologies, particularly Convolutional Neural Networks (CNNs) [5] These sophisticated models, capable of autonomously learning and extracting salient features from image data, have significantly outperformed traditional approaches, leading to substantial advancements in the accuracy and reliability of emotion detection systems. Furthermore, contemporary research in this domain explores integrating attention mechanisms and advanced memory-based architectures, such as Long Short-Term Memory (LSTM)[6] networks, to refine FER systems' performance further.

A critical aspect of FER systems is the feature selection process, which involves identifying the most relevant and informative features from the data. This process is pivotal for the performance of emotion recognition models. Recent developments in quantum computing present promising avenues for enhancing feature selection methods, offering the potential for significant improvements in both accuracy and computational efficiency.

The focus of this research is twofold: firstly, to design an innovative Deep Neural Network (DNN) architecture that leverages the combined strengths of Local Binary Patterns (LBP) [7] and Histogram of Oriented Gradients (HOG) features (HOG) [8]; secondly, to introduce the integration of Gazelle optimization [9], a novel swarm optimization metaheuristic, with quantum computing principles, for optimizing feature selection in the proposed DNN architecture. The primary contribution of this study is the demonstration of quantum computing's potential to significantly enhance the feature selection process, yielding improved accuracy and efficiency in emotion recognition systems compared to traditional methodologies.

The structure of this paper is as follows: Section 2 details the proposed methodology, incorporating Gazelle optimization and Quantum theory for optimized feature selection. Experimental setup and results are presented in Section 3, followed by a thorough discussion and analysis of the findings. The paper concludes with Section 4, summarizing key insights and delineating potential avenues for future research in this field, emphasizing the implications and possibilities afforded by quantum computing and swarm optimization in the realm of feature selection for emotion recognition systems.

2. Proposed approach

Fig.1 provides a comprehensive presentation of the proposed approach. The steps involved in the study will be detailed in the following, starting from the pre-processing of the data and ending with the evaluation of the results.

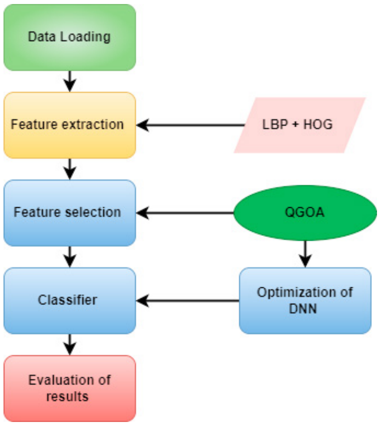

Fig. 1. QGOA Flowchart

2.1. Feature extraction

In this paper, we enhance feature extraction by integrating Histograms of Oriented Gradients (HOG)[8] and Local Binary Patterns (LBP)[7], utilizing their complementary strengths. LBP excels in capturing local texture details such as wrinkles, while HOG effectively delineates the shape and structure of faces. This combination not only provides a richer representation of facial expressions by encompassing both texture and structural details but also improves robustness against variations in lighting, pose, and limited data intensity. This approach ensures a more comprehensive and resilient feature representation, crucial for robust facial expression recognition[10].

2.2. Feature selection

The goal of feature selection is to identify the optimal subset of features that minimizes classification error and the number of features used [11, 12]. This process uses a fitness function evaluated through the K-Nearest Neighbor (KNN) classifier, as detailed in Eq.(1). The fitness function accounts for the classification error rate, denoted as $\alpha\gamma_R(D)$, where $\gamma_R(D)$ is the error rate of the classifier R for decision D, and α modifies the impact of classification quality on the overall fitness. Additionally, the function considers the proportion of selected features, expressed as $\beta\frac{|R|}{|C|}$, where β adjusts the emphasis on reducing the feature count relative to the total number of features available [13]. This dual focus ensures a balanced feature efficacy and efficiency assessment in the selection process.

$$\downarrow \text{ fitness } = \alpha\gamma_R(D) + \beta\frac{|R|}{|C|} \tag{1}$$

The number of selected features, $\beta\frac{|R|}{|C|}$, is represented by the length of the selected feature subset, $|R|$, divided by the total number of features, $|C|$. The parameter $\beta = (1 - \alpha)$ regulates the significance of the subset length compared to the overall fitness. According to [14], α and β are two parameters that correspond to the importance of classification quality and subset length.

2.3. Gazelle Optimization

The Gazelle Algorithm [9] is an optimization method inspired by the survival strategies of gazelles. It enhances robustness and efficiency in solving engineering design problems by emulating gazelles' grazing and evasion tactics.

2.3.1. GOA Formulation

- **Population initialization** GOA is a population-based optimization algorithm that uses randomly generated gazelles as search agents to find the optimal solution. The gazelles represent candidate solutions as an n-dimensional matrix in the search process Eq.(2).

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,d} \\ x_{2,1} & x_{2,2} & \dots & x_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \dots & x_{n,d} \end{bmatrix} \tag{2}$$

  In the model, 'n' represents the number of gazelles (search agents), and 'd' is the number of dimensions in the problem space. Each '$x_{i,j}$' denotes the j^{th} component of the i^{th} gazelle's candidate solution.
  Candidate positions 'x_{ij}' are randomly established within defined upper (UB) and lower (LB) bounds according to equation (3). The "best gazelle" is identified and forms the Elite matrix, influencing the next iteration's search. This process allows for systematically exploring the bounds to discover the optimal solution.

$$x_{i,j} = \text{rand} \times (UB_j - LB_j) + LB_j \tag{3}$$

  In every iteration, each element of the matrix X generates a candidate solution. The solution with the lowest value is considered the best solution and used to search for the next step in the algorithm. Elite matrix contains the nominated best candidate which guides the search process for future solutions, is defined as follow Eq.(4):

$$\text{Elite} = \begin{bmatrix} x'_{1,1} & x'_{1,2} & \dots & x'_{1,d} \\ x'_{2,1} & x'_{2,2} & \dots & x'_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ x'_{n,1} & x'_{n,2} & \dots & x'_{n,d} \end{bmatrix} \tag{4}$$

  A random search agent motion characterized by a displacement that follows the normal(Gaussian) distribution function with a mean $\mu = 0$ and a unit variance $\sigma = 1$. It is determined in Eq.(5)

$$\begin{aligned} f_B(x;\mu,\sigma) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \end{aligned} \tag{5}$$

  The Lévy flight is characterized by a random walk pattern that adheres to the Lévy distribution, which possesses a power-law tail.

$$L\left(x_j\right) \approx \left|x_j\right|^{1-\alpha} \tag{6}$$

  Equation (7) represents the Lévy stable process as an integral form.

$$f_L(x;\alpha,\gamma) = \frac{1}{\pi} \int_0^\infty \exp\left(-\gamma q^\alpha\right) \cos(\text{qx}) \delta q \tag{7}$$

- **Exploitation** In this phase, gazelle agents forage calmly, unaffected by predators. They navigate local space efficiently using standard Brownian motion (Eq.(5)), characterized by uniform and systematic steps, to simulate their grazing movements. The mathematical model of this behaviour is expressed in Eq.(8).

$$\begin{aligned} \text{gazelle}_{\,i+1} &= \text{gazelle}_{\,i} + s * \vec{R} * \vec{R}_B \\ &\quad * \left(\text{Elite}_{\,i} - \vec{R}_B * \text{gazelle}_{\,i}\right) \end{aligned} \tag{8}$$

  In this model, the next iteration solution, $gazelle_{i+1}$, is calculated from the current solution $gazelle_i$. The grazing speed is denoted by s, while the vectors $\vec{R}$ and $\vec{R}_B$, containing random and uniform random numbers respectively, represent Brownian motion and are bounded between 0 and 1.

- **Exploration** The exploration phase initiates when a predator is detected. The mathematical representation of the Gazelle's behaviour when it spots the predator is described by Eq.(9).

$$\overrightarrow{\text{gazelle}}_{i+1} = \overrightarrow{\text{gazelle}}_i + S \cdot \mu \cdot \text{CF} * \cdot \vec{R}_B * \cdot \left(\overrightarrow{\text{Elite}}_i - \vec{R}_L * \cdot \text{gazelle}_i \right) \tag{9}$$

Equation (10) presents the mathematical model for the predator's behaviour chasing the Gazelle.

$$\overrightarrow{\text{gazelle}}_{i+1} = \overrightarrow{\text{gazelle}}_i + S \cdot \mu \cdot \text{CF} * \cdot \vec{R}_B * \cdot \left(\overrightarrow{\text{Elite}}_i - \vec{R}_L * \cdot \overrightarrow{\text{gazelle}}_i \right) \tag{10}$$

where CF denotes the cumulative effect of the predator Eq.(11)

$$CF = \left(1 - \frac{\text{iter}}{\text{Max iter}}\right)^{\left(2 \frac{\text{iter}}{\text{Max} - \text{ier}}\right)} \tag{11}$$

The survival rate of 0.66 implies that predators have a success rate 0.34 in capturing gazelles. This Predator Success Rate (PSR) is integral to the algorithm, ensuring it avoids becoming trapped in sub-optimal solutions or local minima. The influence of PSR on the algorithm is mathematically formulated in Eq.(12).

$$\overrightarrow{\text{gazelle}}_{i+1} = \begin{cases} \overrightarrow{\text{gazelle}}_i + \text{CF}[\overrightarrow{\text{LB}} + \vec{R} \cdot (\overrightarrow{\text{UB}} - \overrightarrow{\text{LB}})] \cdot \vec{U}, & \text{if } r \leq \text{PSRs} \\ \overrightarrow{\text{gazelle}}_i + [\text{PSRs} \cdot (1 - r) + r] \cdot (\overrightarrow{\text{gazelle}}_{r_1} - \overrightarrow{\text{gazelle}}_{r_2}), & \text{else} \end{cases} \tag{12}$$

2.3.2. *Quantum-based Gazelle Optimization Algorithm: QGOA*

Several metaheuristics have incorporated quantum theory to tackle optimization challenges beyond the reach of traditional algorithms [15]. Quantum theory introduces concepts like quantum parallelism and quantum tunnelling into optimization algorithms. Quantum parallelism allows the simultaneous exploration of multiple solutions, accelerating the search process [16]. Quantum tunnelling permits exploration across significant energy barriers in the search space, potentially overcoming local minima to find the optimal solution. Consequently, we advocate for a quantum-enhanced version of the GOA algorithm to devise more effective methods for binary optimization problems.

Quantum Representation : Each quantum entity q is associated with a phase vector q_θ, represented by a sequence of phase angles θ_i, where $1 \leq i \leq m$. This can be expressed as:

$$q_\theta = [\theta_1, \theta_2, \ldots, \theta_m], \theta_i \in [0, 2\pi] \tag{13}$$

where m is the number of qubits in each quantum entity q, each quantum entity q is represented by a string of qubits.

$$q = \left[\begin{array}{c|c|c|c} \cos(\theta_1) & \cos(\theta_2) & \cdots & \cos(\theta_m) \\ \sin(\theta_1) & \sin(\theta_2) & & \sin(\theta_m) \end{array} \right] \tag{14}$$

A pair of numbers represents the probability amplitudes of a quantum bit $(\cos(\theta_i), \sin(\theta_i))$. The probability of choosing item x_i is given by $|\sin(\theta_i)|^2$, while the probability of rejecting item x_i is expressed as $|\cos(\theta_i)|^2$.

The quantum population at the t^{th} generation, $Q(t)$, is composed of the quantum individuals and has a size of n individuals, represented as $Q(t) = \left(q_1^t, q_2^t, \ldots, q_n^t\right)$.

Position Update : The Quantum Rotation Gate (QRG) strategy [17] used in the QGOA enhances the exploitation phase by updating the positions of search individuals. In quantum computing, the QRG processes states and converts the binary quantum bits from the swarm-based algorithm into continuous data. This involves updating the dimensions of the search agent's information in pairs using a QRG, which rotates the information, represented as a 2x2 matrix in equation (15).

$$U(\theta_i) = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \end{bmatrix} \tag{15}$$

Therefore, the Q-bit individual string and position of the quantum gazelles are updated as described in Eq. (16).

$$\begin{bmatrix} \alpha_i(t+1) \\ \beta_i(t+1) \end{bmatrix} = U(\theta_i) \times \begin{bmatrix} \alpha_i(t) \\ \beta_i(t) \end{bmatrix}, \tag{16}$$

In the context of our study, feature selection is treated as a binary optimization problem where each feature is either selected (1) or not selected (0). The QGOA employs a quantum-inspired approach to generate and optimize feature subsets, ultimately determining the most effective combination of features for improving the performance of our Deep Neural Network (DNN) model in facial emotion recognition tasks.

Equation (17) plays a pivotal role in the final stage of the QGOA process, where the probabilistic representation of features, influenced by quantum mechanics, is converted into a definitive binary decision. Specifically, the equation is formulated as follows:

$$x_i^t = \begin{cases} 1, & \text{if } r < |\beta_i(t+1)|^2, \\ 0, & \text{otherwise}, \end{cases} \tag{17}$$

where, x_i^t represents the binary state of the i-th feature at iteration t, and $\beta_i(t+1)$ denotes the probability amplitude associated with the i-th feature being selected in the next iteration $t+1$. The random number is referred to as r. Complex numbers α and β have probability amplitudes of $|\alpha|^2$ and $|\beta|^2$, respectively.

The binary transformation achieved by Equation (17) is crucial because it allows the algorithm to decide definitively which features to include in the model based on the quantum probability estimates. This approach harnesses the stochastic nature of quantum mechanics to explore a wide range of possible feature combinations more efficiently than classical algorithms, potentially escaping local minima and converging towards a more optimal subset of features. By transforming these probabilities into binary decisions, we effectively apply a quantum-inspired thresholding mechanism that determines the final selection of features used by the DNN for facial emotion recognition. This method enhances the feature selection process by allowing for a dynamic exploration of the feature space, guided by both the probabilistic tendencies of quantum states and the deterministic needs of machine learning classification tasks.

Solution Space Transformation : The evaluation of each individual's performance within the algorithmic framework is quantified using a fitness value, serving as a metric to gauge the quality level of each worker. A pivotal aspect of this process is the transformation of the individual's position within the solution space. The designated solution space for the given problem is denoted by $\Omega = [a, b]$. The mechanism for altering the solution space is encapsulated in the subsequent equations:

$$\begin{aligned} RG_{ic} &= \frac{a(1-\alpha_i) + b(1+\alpha_i)}{2} \\ RG_{is} &= \frac{a(1-\beta_i) + b(1+\beta_i)}{2} \end{aligned} \tag{18}$$

The pseudo-code of the Quantum-Gazelle optimization algorithm is described in Algorithm 1.

3. Experimental results and discussions

3.1. Dataset description

Datasets are crucial for training and validating machine learning algorithms. It functions as a compilation of crucial characteristics for recognizing facial emotions. FER2013 [18] is a popular dataset for emotion recognition introduced at ICML 2013. It contains 35,887 grayscale images of faces in 48x48 size, each labelled with one of seven emotions: anger, neutral, disgust, fear, happiness, sadness, or surprise. The images have the faces occupying most of the space for easier feature extraction. Fer2013 is split into a training set with 28,709 examples and a test set with 3,589 examples.

Algorithm 1 Pseudo code of Quantum-GOA

Initialize the algorithm parameters $(s, \mu, S, R, r, PSR, max_{iter}, \ldots)$
Initialize the gazelle populations

while $iter \leq max_{iter}$ **do** Binary transformation using Equation (17). (Only applied for binary problems (i.e. FS problem))
Calculate the fitness of the gazelles.
top-gazelle $\leftarrow fittestgazelle$
Construct the Elite gazelle matrix

  if $r1 \leq 0,8$ **then**

    if $r2 \leq 0,5$ **then** Exploitation: Apply Equation (8).

    elseExploration:

      if mod(iter, 2) == 0 **then** $\mu = -1$

      else$\mu = 1$

        if iter < size(gazelle, 1) / 2 **then** Update First N/2 gazelles based on Equation (9).

        elseUpdate Last N/2 gazelles based on Equation (10).

        end if

      end if

    end if

Apply Quantum rotation gate using Equation (16).
Transform Solution space using equation (18).
Fitness update
top-gazelle update
Elite update
Applying PSRs effect and update based on Equation (12).
Return top-gazelle

3.2. DNN hyperparamters

Hyperparameter optimization is crucial in training Deep Neural Networks (DNNs) as it affects the network's performance, speed, accuracy, and ability to generalize to new data [19]. Optimal hyperparameters can lead to better performance, faster convergence, better generalization, and improved resistance to overfitting. Thus, carefully selecting the correct hyperparameters can improve the network's performance and make its training more efficient. Table 1 presents the range of parameters utilized to find the most appropriate architecture of our Deep Neural Network (DNN) for facial emotion classification.

Table 1. Node and global hyperparameters

Node hyperparameter	***Range***
Number of Filters	[32, 256]
Activation function	ReLu,sigmoid
Global hyperparameters	***Range***
Learning Rate	[0,0001 , 0,1]
batch size	[20,50]
Number of hidden layers	[2,4]
Number of neurons	[100,200]
Dropout rate	[0.2,0.5]

3.3. Results

During these simulations, we set the number of maximum iterations to 100 and the size of the population of each MH technique to 30. To ensure a robust assessment of the proposed approach, we employed a 10-fold cross-validation technique. The use of 10-fold cross-validation in our study not only enhances the reliability of our results but also ensures that the evaluation is thorough and indicative of the proposed method's effectiveness across various scenarios. This evaluation begins with a comparison of various MH-based techniques. Therefore, we selected to compare our proposed approach, using QGOA, with three others: Gazelle Optimization Algorithm (GOA), Quantum Particle Swarm Optimization (QPSO) [20], and Quantum Genetic Algorithm (QGA) [21].

Table 2. Combined Comparison of FS Techniques with Different Classifiers and Feature Extraction Methods

FS Technique	Feature Extraction	Classifier	Accuracy	Precision	Recall	Auc	F1_Score
QGOA	LBP+HOG	DNN_QGOA	**0.93257**	**0.88422**	**0.93295**	**0.96086**	**0.90174**
GOA	LBP+HOG	DNN_GOA	0.87225	0.81043	0.8716	0.92521	0.82408
QPSO	LBP+HOG	DNN_QPSO	0.90917	0.85627	0.90135	0.94308	0.87239
QGA	LBP+HOG	DNN_QGA	0.90164	0.84632	0.90301	0.94332	0.86412
QGOA	Inception V3	DNN_QGOA	0.92561	0.87703	0.92756	0.95758	0.89449
QGOA	ResNet50	DNN_QGOA	0.90833	0.85075	0.9003	0.94253	0.86706
QGOA	DenseNet201	DNN_QGOA	0.92045	0.86784	0.92134	0.95406	0.88584

Table 2 provides a comparative analysis of different Feature Selection (FS) techniques, each paired with distinct Feature Extraction methods, evaluated across a spectrum of performance metrics including Accuracy, Precision, Recall, AUC (Area Under the Curve), and F1_Score. These metrics are essential for assessing the efficacy of facial expression recognition systems. The FS techniques under consideration are Quantum Gazelle Optimization Algorithm (QGOA), Gazelle Optimization Algorithm (GOA), Quantum Particle Swarm Optimization (QPSO) [20], and Quantum Genetic Algorithm (QGA) [21]. Feature extraction methodologies encompass combinations of Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), as well as advanced neural network architectures like ResNet50 [26], Inception V3 [27] and DenseNet201 [28]. Notably, the DNN_QGOA classifier is utilized across all configurations, with specific FS approaches paired accordingly.

The analysis reveals that the integration of QGOA with LBP and HOG for feature extraction yields the most superior outcomes across the board. This suggests that the optimization capacity of QGOA significantly enhances the classifier's ability to accurately predict facial expressions by refining the selected features. Conversely, other FS techniques employing LBP and HOG manifest a slight decrement in performance relative to QGOA, indicating the pivotal role of the optimization process in maximizing the classifier's efficacy.

Moreover, the application of QGOA alongside Inception V3, ResNet50, and DenseNet201 illustrates competitive but somewhat lower performance compared to the traditional LBP+HOG combination. Among these, Inception V3 emerges as the most effective feature extraction method when optimized with QGOA, highlighting the potential of integrating QGOA with deep learning-based feature extraction to improve facial expression recognition systems. However, the slightly diminished performance compared to the LBP+HOG method optimized by QGOA suggests that the complexity of features extracted by these advanced architectures might necessitate alternative optimization strategies or adjustments in the model configurations.

Table 3 presents the optimized hyperparameters for a deep neural network, determined using the Quantum Gazelle Optimization Algorithm (QGOA), categorized into node-specific and global settings. It outlines optimal values for parameters such as the number of filters, activation functions, learning rate, batch size, and more, tailored for facial expression recognition tasks.

Table 3. Optimized hyperparameters using QGOA algorithm

Node hyperparameter	***Range***
Number of Filters	[79]
Activation function	ReLu
Global hyperparameters	***Range***
Learning Rate	0,0009
batch size	[24]
Number of hidden layers	3
Number of neurons	178
Dropout rate	0.46

3.4. Detailed Analysis of Experimental Results

Our examination of the experimental results reveals the mechanisms driving the superior performance of the proposed method, which integrates the Quantum Gazelle Optimization Algorithm (QGOA) with traditional feature extraction techniques—Local Binary Patterns (LBP) and Histograms of Oriented Gradients (HOG)—alongside an optimized Deep Neural Network (DNN). This combination enhances the facial emotion recognition system significantly, as evidenced by comparative analyses with other feature selection techniques, showcasing the distinct advantages of the QGOA in this domain. Firstly, the QGOA enhances the feature selection process by pinpointing the most informative and discriminative features extracted via LBP and HOG. Leveraging quantum-inspired mechanisms allows for a deeper exploration of the feature space, which traditional methods may not adequately cover. This capability is crucial in FER, where the subtle nuances in facial expressions demand precise feature discrimination. The optimization selects and arranges the features to maximize their informational value for the classification tasks undertaken by the DNN.

Our method, powered by QGOA, optimizes the DNN architecture by adjusting hyperparameters such as the number of filters, layers, and learning rates. These adjustments are pivotal for the network's ability to learn efficiently from complex datasets. The optimal configuration of these parameters, determined by QGOA, helps balance the model's capacity to generalize across different emotional expressions while avoiding overfitting. This fine-tuning is essential for enhancing the model's accuracy and robustness in practical scenarios. The performance metrics underscore the benefits of using QGOA. The classifier, when enhanced with QGOA, achieves higher accuracy, precision, recall, and F1 scores than other methods like GOA, QPSO, and QGA. This improvement is attributed to the efficient handling of both feature selection and network optimization within a unified framework. The high AUC value further indicates the model's effectiveness in distinguishing between different emotional states, which is a critical capability in FER applications.

4. Conclusions and future works

The work presented in this paper constitutes a notable advancement in facial expression recognition, highlighting the effectiveness of a hybrid deep-learning framework. This study ingeniously combines Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) for initial feature extraction, utilizing the strengths to capture a broad array of facial features. The foundation is optimized using the novel Quantum Gazelle Optimization Algorithm (QGOA), significantly boosting the deep neural network's (DNN) performance in the classification phase. Experimental results confirm the QGOA's superiority in feature optimization, leading to improved recognition rates.

Future research could address more complex detection scenarios involving eyeglasses, facial hair, and cosmetics affecting facial expression visibility. Enhancing feature extraction and optimization by integrating advanced image processing techniques or adaptive neural network architectures might be effective. Further refining the QGOA to handle these complexities could involve incorporating contextual information or employing dynamic optimization strategies.

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